

Samuele Ferracin

(631) 697-4519 | samueleferracin@hotmail.com | [Google Scholar](https://scholar.google.com/citations?user=UW8vY8IAAAJ) | [linkedin.com/in/samueleferracin](https://www.linkedin.com/in/samueleferracin) | samuferracin.github.io

Mechanical Engineer PhD candidate specializing in nonlinear FEA, structural dynamics, multiphysics simulation, and experimentally validated hardware design. Experience developing Abaqus, COMSOL, MATLAB models for buckling/post-buckling, dynamic response, fluid–structure interaction, contact, and thermo-electromagnetic systems. Strong hands-on background in CAD, prototyping, 3D printing, and validation of mechanical systems. Available for full-time roles in Bay Area from Nov 2026.

TECHNICAL SKILLS

FEA / CAE: Abaqus, COMSOL, nonlinear Finite Element Analysis, large deformation, buckling/post-buckling, implicit/explicit dynamics, contact/friction, structural analysis, stress analysis, modal analysis, structural dynamics, electro-thermal simulations

Modeling / Programming: MATLAB, Python, reduced-order modeling, parametric studies, data analysis, neural networks

Mechanical Design: SolidWorks, Autodesk Inventor, mechanism design, compliant/multistable structures, material selection, prototype design, design-for-test, technical drawings, geometric dimensioning and tolerancing (GD&T)

Testing / Validation: Wind-tunnel testing, DIC, load-cell measurements, design of experiments, model validation, impact loading

Manufacturing / Prototyping: FDM, SLA, DIW, microscale 3D printing, conductive elastomer composites, soft materials

EDUCATION

University of Pennsylvania

Philadelphia, PA | Sept 2021 - Expected Oct 2026

Ph.D. in Mechanical Engineering

- Research focus: nonlinear FEM, multistable structural dynamics, fluid-structure interaction, experimental validation
- Teaching assistant: Led lectures, recitations, office hours, and grading for undergraduate and graduate engineering students.

Polytechnic University of Turin

Turin, Italy | 2018 - 2021

M.Sc. in Aerospace Engineering - Propulsion Systems

- GPA: 106/110 (~3.85/4.0), Top 5%, merit-based scholarship recipient
- Coursework: propulsion systems, supersonic flow, CFD of turbomachinery, aerospace structures

Polytechnic University of Turin

Turin, Italy | 2015 - 2018

B.Sc. in Aerospace Engineering

- GPA: 106/110 (~3.85/4.0), Top 3% - Honors: “Young Talents” Program (top ~4% admitted) + scholarship
- Coursework: structural mechanics, thermodynamics, aerodynamics, materials science, technical drawing, GD&T

ENGINEERING & RESEARCH EXPERIENCE

University of Pennsylvania, Mechanical Engineering and Applied Mechanics

Philadelphia, PA | Sept 2021 - Present

Ph.D. Student Researcher (Advisor: Prof. Jordan Raney)

- Developed nonlinear Abaqus Finite Element Analysis (FEA) models for multistable mechanical structures including large deformation, buckling and post-buckling, contact, friction, and transient dynamic response; used simulations to guide geometry, material, and contact-interface design decisions.
- Built reduced-order MATLAB models for nonlinear structural dynamics, modal coupling, snap-through, stability, and forced response of bistable beams and plates.
- Designed, fabricated, and tested prototypes and custom experimental setups; validated nonlinear FEM and reduced-order model predictions against measured deformation, force response, and stability transitions.
- Prepared technical presentations, sponsor-review materials, and documentation of simulation assumptions, validation results, and experimental findings.

Selected projects:

- Designed, prototyped, and tested elastomeric multistable metamaterials with contact-based tunable stiffness up to 50x, using parametric FEA to guide geometry and interface design.
- Developed coupled fluid-structure interaction models for bistable plates in unsteady vortex-driven flow; validated predicted modal response using wind-tunnel experiments with CNC-machined supports and PET films, DIC tracking, and load-cell measurements.
- Developed analytical and reduced-order models for phonon-controlled pinning and depinning of nonlinear waves in bistable metamaterials, deriving defect-pinning thresholds and nonlinear resonance for remotely reprogrammable mechanical memory.
- Designed and fabricated bistable SLA-resin 3D-printed vortex generators and elastomeric multistable hinges for flow-separation control, gust-load alleviation, and wing-surface morphing.
- Developed data-driven multiphysics simulations using structure-preserving neural networks to infer nonlinear dynamics from experimental displacement and force data.
- Developed 3D-printed liquid-crystal-elastomer/liquid-metal soft actuators and microscale multistable metamaterials using conductive elastomer composites and pore-scale architectures for robotics, prosthetics, and cell-interaction experiments.

Stony Brook University

Stony Brook, NY | 2020 - 2021

M.Sc. Thesis (Advisor: Prof. Paolo Celli)

- Designed, simulated, and experimentally tested a bistable deployable morphing structure for space applications, focusing on buckling/post-buckling response and vibration-induced shape change.
- Developed Abaqus FEA models to predict stable post-buckling configurations and dynamic response; compared simulation results against prototype experiments.
- Supported by a merit-based research fellowship.

Polytechnic University of Turin, Superconductivity Lab

Turin, Italy | 2018 - 2020

Junior Research Fellow

- Developed COMSOL multiphysics FEA simulations of superconducting bolometer response, coupling superconducting and electrothermal effects.
- Designed and tested experimental configurations to evaluate thermo-electromagnetic behavior.
- Modeled a superconducting magnetic energy storage system (SMES) with ENI S.p.A.

Ladar Ltd.

Oslo, Norway | 2018

Summer Engineering Intern

- Designed a mechanical laser-alignment mechanism for maritime LIDAR sensors in Autodesk Inventor, resolving tight packaging constraints, selecting marine-grade materials to mitigate harsh environmental corrosion.

Tiologic S.r.l.

Solagna, Italy | 2016

Summer Engineering Intern

- Developed custom test fixtures and camera-positioning mechanisms to optimize optical inspection stands, leveraging FDM 3D printing to quickly iterate designs and improve measurement repeatability.

SELECTED PUBLICATIONS

Full publication list on Google Scholar. *Equal contribution.

- **S. Ferracin**, D. Jin, V. Tournat, J.R. Raney. *Phonon controlled mechanical memory via pinning and depinning of transition waves*. Under review at Physical Review B, 2026.
- D. Jin, **S. Ferracin**, V. Tournat, S. Patel, P.K. Purohit, J.R. Raney. *Transition waves in a beam coupled to a foundation of symmetric bistable elements*. Physical Review Applied, 2025.
- A.B.M.T. Haque*, **S. Ferracin***, J.R. Raney. *Reprogrammable mechanics via individually switchable bistable unit cells in a prestrained chiral metamaterial*. Advanced Materials Technologies, 2024.
- A. Rahman, **S. Ferracin**, S. Tank, C. Zhang, P. Celli. *Shape-retaining beam-like morphing structures via localized snap through*. International Journal of Solids and Structures, 2024.
- Q. He*, **S. Ferracin***, J.R. Raney. *Programmable responsive metamaterials for mechanical computing and robotics*. Nature Computational Science, 2024.
- Oral presentation at **SES 2023, SES 2025, FMI Workshop 2026, USNC-TAM 2026**; recipient of American Academy of Mechanics Travel Fellowship.